

Supervised vs Unsupervised Learning

A simple notebook for complete beginners. No code, no maths.

Read this line first. Machine learning means a computer learns patterns from data instead of being told the rules by a programmer. There are two main ways it can learn, and the **ONLY** thing that separates them is one question: **does the data already contain the correct answers?**

1. The one idea you must understand: the label

A **label** is the correct answer attached to a row of data. If you have a photo and someone already wrote "this is a cat" on it, then "cat" is the label. A dataset where every row has a label is called a **labeled dataset**. A dataset with no answers attached is an **unlabeled dataset**.

LABELED data -> Supervised Learning



UNLABELED data -> Unsupervised Learning



Figure 1: The same points. On the left the answers are known. On the right they are not.

2. The two families at a glance

SUPERVISED LEARNING

You give the model questions **and** the correct answers. It studies them, learns the connection, then answers new questions on its own.

Like: a student studying with an answer key.

Needs: a labeled dataset.

Goal: predict the answer for new data.

UNSUPERVISED LEARNING

You give the model data only. No answers. It looks for structure, similarity and patterns hiding inside the data by itself.

Like: a student sorting a pile of books with no instructions.

Needs: an unlabeled dataset.

Goal: discover something you did not know.

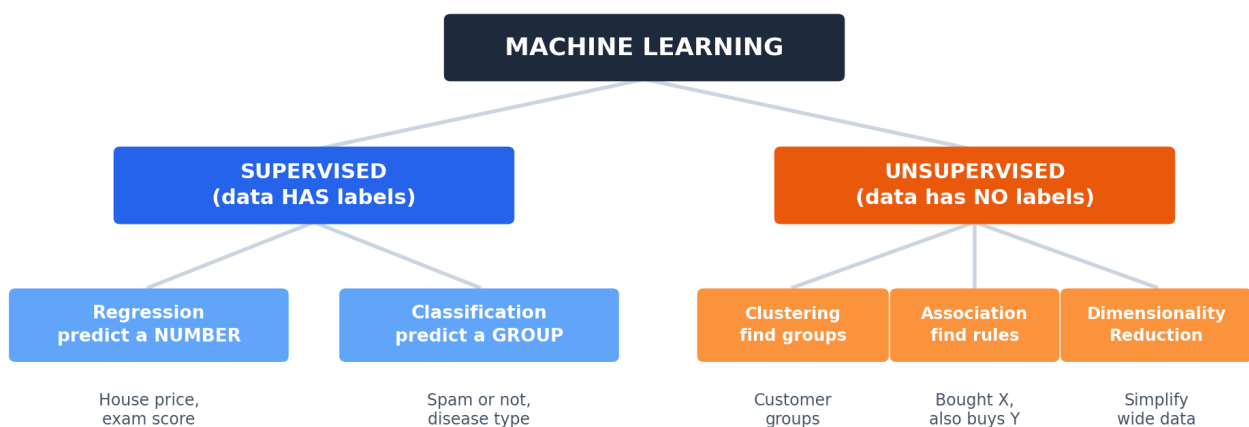


Figure 2: The full family map. Everything in this notebook fits somewhere on this tree.

SUPERVISED LEARNING

3. Definition in plain words

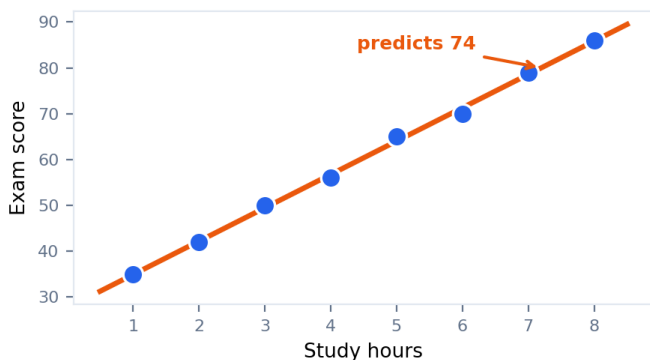
Supervised learning is teaching by example. You show the computer many solved examples. Each example has some **input columns** (called features) and one **answer column** (called the label or target). The computer keeps guessing, checks its guess against the real answer, sees how wrong it was, and adjusts itself. After enough examples it can handle inputs it has never seen before.

How it works, step by step

- 1 **Collect labeled data.** Rows of inputs where the correct answer is already known.
- 2 **Split the data.** Most rows for training, a small part kept hidden for testing.
- 3 **Train the model.** It compares its guess to the real answer and reduces the error.
- 4 **Test it.** Use the hidden rows to check if it really learned or just memorised.
- 5 **Use it.** Feed in new data with no answer and let it predict.

4. The two types of supervised learning

REGRESSION: draw a line, predict a number



CLASSIFICATION: draw a border, predict a group

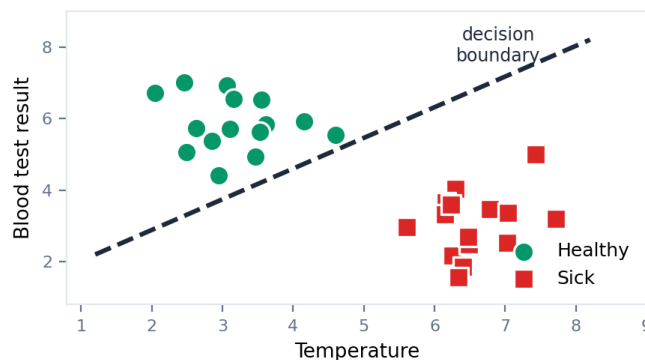


Figure 3: Regression fits a line through the points. Classification draws a border between groups.

TYPE 1: REGRESSION

The answer is a **number on a scale**. It can be any value in a range, and values close to each other mean similar things.

Question it answers: how much? how many?

Answers look like: 74.5, 1200, 38.2

Common models: Linear Regression, Decision Tree Regressor, Random Forest Regressor

TYPE 2: CLASSIFICATION

The answer is a **category from a fixed list**. There is no order and no in between value.

Question it answers: which one? yes or no?

Answers look like: Spam, Malaria, Approved

Common models: Logistic Regression, Decision Tree, Random Forest, Naive Bayes

SUPERVISED LEARNING - SAMPLE DATASETS

Dataset A - Student performance (REGRESSION)

The green column is the label. It holds a number, so this is a regression problem.

Student	Study hours per week	Attendance %	Past average	LABEL: Exam score
S001	2	60	41	45
S002	5	78	58	62
S003	8	95	74	84
S004	3	70	50	52
S005	10	98	88	93
S006	6	85	66	71
S007 (new)	7	90	70	= ? to predict

What the model learns here: more hours plus higher attendance usually means a higher score. For student S007 it would predict roughly 78. Notice the answer is a number that could also have been 77.4 or 79.1.

Applications of this kind of dataset

Where it is used	Input columns become	The predicted number is
School and university	Study hours, attendance, past marks	Final exam score
Housing market	Size, rooms, location, age of building	Price of the house
Hospital planning	Age, symptoms, test values, admission day	Days until patient recovers
Shops and business	Season, past sales, price, advertising spend	Sales for next month

Dataset B - Patient screening (CLASSIFICATION)

Same idea, but now the green label column holds a category, not a number.

Patient	Fever	Cough	Temp (C)	Age	LABEL: Diagnosis
P001	Yes	Yes	39.2	24	Malaria
P002	No	No	36.8	31	Healthy
P003	Yes	No	38.5	45	Infection
P004	Yes	Yes	39.6	8	Malaria
P005	No	Yes	37.1	52	Common cold
P006	No	No	36.6	19	Healthy
P007 (new)	Yes	Yes	39.0	27	= ? to predict

What the model learns here: fever plus cough plus high temperature points to Malaria, so for P007 it predicts "Malaria". The answer must be one of the words already seen in the label column.

Applications of this kind of dataset

Where it is used	Input columns become	The predicted category is
Health facilities	Symptoms, temperature, age, test results	Disease type or Healthy
Email systems	Words used, sender, links inside	Spam or Not spam
Banks	Amount, location, time, spending history	Fraud or Genuine
Loan offices	Income, job type, repayment history	Approve or Reject

UNSUPERVISED LEARNING

5. Definition in plain words

Unsupervised learning is learning by observation. Nobody tells the computer what is right or wrong, because there is no answer column at all. The computer measures how similar the rows are to each other and organises them. It cannot tell you what a group means, only that a group exists. **A human still has to look at the result and give it a meaning.**

How it works, step by step

- 1 **Collect data with no labels.** Just the input columns.
- 2 **Choose what similar means.** Usually how close two rows are when you compare their numbers.
- 3 **Let the model organise.** It groups similar rows or finds repeating combinations.
- 4 **A human reads the result.** You look at each group and decide what to call it.
- 5 **Act on it.** Use the discovered groups for decisions, or as input to other work.

6. The three types of unsupervised learning

Type	What it does	Everyday example	Common models
Clustering	Puts similar rows into groups	Split customers into buyer types	K-Means, Hierarchical, DBSCAN
Association	Finds things that appear together	People who buy bread also buy milk	Apriori, FP-Growth
Dimensionality reduction	Shrinks many columns into a few, keeping the meaning	Turn 60 survey questions into 3 scores	PCA, t-SNE

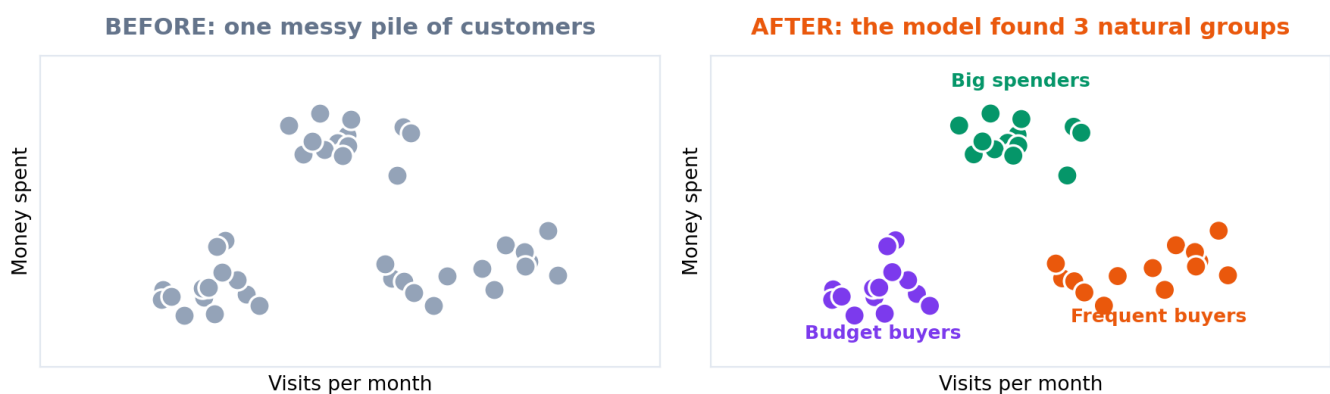


Figure 4: Clustering in action. Nothing was labeled. The model only saw that some customers behave alike.

Dataset C - Customer behaviour (CLUSTERING)

Look carefully: there is **no answer column**. That is the whole point.

Customer	Visits per month	Money spent	Items per visit	Answer column
C001	12	480	9	none
C002	2	35	2	none
C003	11	510	8	none
C004	3	40	3	none
C005	5	620	14	none

What the model finds here: three natural groups. C001 and C003 come often and spend medium amounts. C002 and C004 come rarely and spend little. C005 comes rarely but spends a lot. The model does not name the groups. **You** name them: frequent buyers, budget buyers, big spenders.

ASSOCIATION, COMPARISON AND RECAP

Dataset D - Shopping baskets (ASSOCIATION)

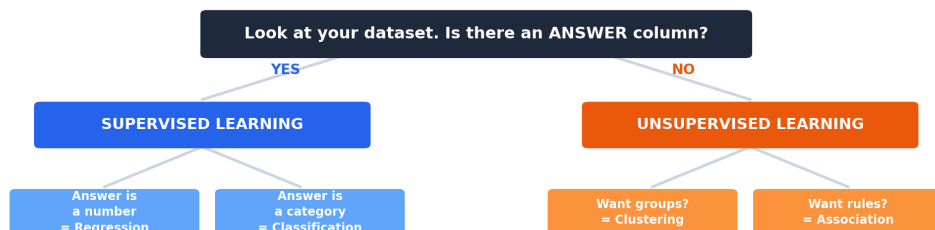
Receipt	Items bought together
T001	Bread, Milk, Eggs
T002	Bread, Milk
T003	Bread, Milk, Sugar
T004	Rice, Oil
T005	Bread, Milk, Tea

What the model finds here: a rule. Bread appears in 4 receipts, and in all 4 of them Milk appears too. So it learns: if a person buys Bread, they very likely buy Milk. **Applications:** shop shelf arrangement, "customers also bought" suggestions online, bundle offers, and in health data, finding symptoms or drugs that keep appearing together.

7. Side by side comparison

Point of difference	Supervised Learning	Unsupervised Learning
Dataset needed	Labeled (inputs + answers)	Unlabeled (inputs only)
Who provides the answer	A human labels it in advance	Nobody
Main goal	Predict the answer for new data	Discover hidden structure
Types	Regression, Classification	Clustering, Association, Dim. reduction
Is the result checkable	Yes, compare prediction to the real answer	Not directly, a human must judge it
Cost of getting data	High, labelling takes people and time	Low, raw data is usually enough
Example in one line	Predict if a patient has malaria	Discover that patients fall into 3 profiles

8. How to decide which one you need



9. Mistakes beginners usually make

The confusion	The correction
Unsupervised learning needs no data	It needs plenty of data. It only needs no labels .
Classification and clustering are the same	Classification puts data into groups you already defined. Clustering discovers groups nobody defined.
Numbers in the label mean regression, always	Not always. If the number is a code such as 1 = Yes and 0 = No, it is a category, so it is classification.
Unsupervised learning is the smarter one	Neither is smarter. They answer different questions. Your data decides which one you can use.

Remember only this if you forget everything else.

Open your dataset and look for a column holding the correct answer.

It is there and it is a number -> supervised, regression.

It is there and it is a category -> supervised, classification.

It is not there -> unsupervised. Group the rows or find rules between them.